



Nature-Inspired Computation

On Campus Exam	Closed Book
Additional Materials: None	Permitted Materials: Approved Non-Programmable Scientific Calculators Permitted (Green)

Guidelines:

- The **duration** of this paper is 2hr
 - Word Count: No Word Count
 - Write all answers in the answer booklet provided.
 - Do not open the exam paper until instructed by the Invigilator.
 - Use only black pen or pencil.
 - Mobile and electronic devices are not allowed.
 - Do not communicate with other candidates during the exam.
-

Exam Instructions:

Answer ALL Questions

Question 1

- (a) Compare and contrast *exploration* and *exploitation* in evolutionary algorithms. How do different *selection* methods affect the balance between these two aspects?

(5 marks)

- (b) Explain how the following components of a genetic algorithm affect its performance:

- Population size
- Mutation rate
- Crossover operator
- Selection pressure

Provide specific *examples* for each component.

(8 marks)

- (c) In the context of Genetic Programming, explain the key differences between program evolution and traditional genetic algorithms. Include a discussion of *representation* and *genetic operators*.

(7 marks)

- (d) State why Boids are good choice for producing a computational model of a flock of birds, and state the three properties of motion used to control the movement of Boids.

(4 marks)

- (e) Explain the role of *velocity* in particle swarm optimisation.

(5 marks)

- (f) Explain why NSGA-II requires both non-dominated sorting *and* crowding distance.

(4 marks)

- (g) The ability to *generalise* and *graceful degradation* are biological principles exhibited by neural networks. Explain what is meant by each.

(4 marks)

ECM3412-RWA

- (h) In the context of cellular automata, explain the properties of *localism*, *parallelism*, and *homogeneity*.

(3 marks)

(Total 40 marks)

Question 2

Consider optimizing a water distribution network using Ant Colony Optimization (ACO). The problem consists of selecting pipe diameters for a network of water pipes.

Problem Specifications:

- Each pipe in the network must be assigned exactly one diameter
- Available diameters: {100mm, 150mm, 200mm, 250mm, 300mm}
- Cost for each pipe = selected diameter $\times$ pipe length
- When water flows through the network, each node must maintain a minimum pressure of 15.0 units
- A network simulator is available that can calculate pressure at each node (connection between two pipes) for any given pipe configuration

For example, if we have a pipe of length 10 meters and choose a diameter of 200mm, its cost would be 2000 units.

- (a) Draw and explain the construction graph that ants would traverse to solve this problem using ACO. How is this different from the TSP construction graph?

(8 marks)

- (b) Design a suitable objective function that will:

- Minimize the total network cost
- Ensure all nodes meet the minimum pressure requirement
- Guide ants towards feasible solutions

(8 marks)

- (c) Design a suitable heuristic function (η_{ij}) to guide ants in selecting pipe diameters. Explain how this heuristic relates to the problem objectives.

(6 marks)

ECM3412-RWA

(d) Specify the following ACO parameters for solving this problem:

- Number of ants
- Pheromone evaporation rate (ρ)
- Relative importance of pheromone trails (α)
- Relative importance of heuristic information (β)
- Initial pheromone values
- Stopping criterion

Justify your choices.

(8 marks)

(Total 30 marks)

Question 3

(a) Particle swarm optimisation has a number of parameters that control the effectiveness of the algorithm.

(i) State two of them and explain their purpose.

(4 marks)

(ii) Suggest an experiment that will confirm that the correct parameters have been chosen.

(5 marks)

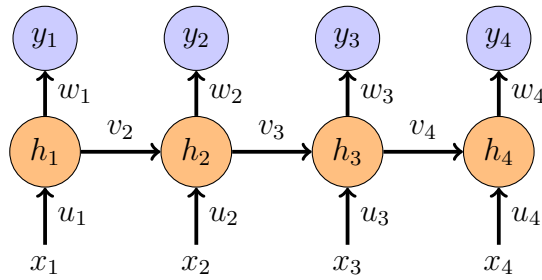
(b) A dataset has 5,000 attributes. Design a multi-objective optimisation problem that identifies which of the attributes are most vital to a given prediction task (the specific prediction task is not important to this question).

Your problem must comprise two objectives. State the objectives, whether they are to be maximised or minimised, and explain why this objective is important to the problem. Outline the solution representation explaining why it is a sensible choice for this problem.

(9 marks)

(c) The diagram below shows a specific type of neural network. State the type of neural network, provide a use case for this specific type of neural network, and explain why it is a sensible choice for this use case.

ECM3412-RWA



(6 marks)

- (d) The Iris dataset contains 150 observations of measurements describing iris flowers. Each flower is described by four features: petal length and width, and sepal length and width. The data contains three classes: *iris-virginica*, *iris-versicolor* and *iris-setosa*. Provide a labelled diagram showing the smallest self-organising map needed to model this data, and explain why your proposed structure is appropriate.

(6 marks)

(Total 30 marks)